

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

Expected May 2028

College Park, MD

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Writing backtested Java capacity-planning logic that sets upper and lower bounds on the EC2 capacity Amazon Redshift reserves for patching to balance availability against cost

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Split a Claude skill into sub-skills backed by Python search and validation tools to stop open-ended model search, which cut LLM workflow generation time **65%** to under 15s with no quality loss in stress tests
- Precomputed popular workflows from a 270K+ row PostgreSQL table into a Redis-cached category tree to keep lookups within the database's permission and rate limits, which became the Claude skill's main search tool
- Rebuilt a legacy test-data workflow tool as a React and TypeScript Electron app with a visual React Flow editor and Jest-tested validation to make editing smoother, which **90%** of 30+ surveyed developers preferred

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Built an EventBridge-scheduled LLM-as-judge check for a production Amazon Bedrock classifier that rates research documents by importance to trigger fine-tuning only when its grades shift past a threshold
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Co-built a topic-scoring research tool and deployed it to dev on AWS Fargate with GitHub Actions CI/CD and AWS CDK to show Amazon Leo's public-policy staff which research topics are popular

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- Built and prompt-tuned a pipeline for a PhD-led robotics project aimed at ICLR that overlays AnyGrasp's candidate grasps on camera images in Open3D to help a vision-language model pick grasps that avoid hitting the table

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Building RAG meal suggestions for the official UMD app with SentenceBERT vector search in ChromaDB to feed GPT-4o mini real scraped recipes, which internal testers rated satisfactory for **93%** of meals
- Built Redis caching in front of Tortuga's Prisma ORM queries to reduce database load and latency for a class scheduler that drew 20,000+ visits in its first 30 days

Projects

IEX Order Book & Matching Engine | *C++17, STL, POSIX, Linux perf*

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine to replay **50.9M** real IEX exchange messages without ever resyncing, which agreed with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices

Technical Skills

AI/ML: LLMs (Claude, Amazon Bedrock, GPT-4o mini), LLM agents and tool use (Claude skills), retrieval-augmented generation (RAG), embeddings and vector search (SentenceBERT, ChromaDB), LLM-as-a-judge evaluation, model fine-tuning, vision-language models, prompt engineering, Open3D

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL

Cloud & Data: AWS (Lambda, Kinesis, EventBridge, Fargate), AWS CDK, monitoring (CloudWatch, X-Ray), REST APIs, PostgreSQL, Redis, OpenSearch

Engineering: full-stack apps (React, Electron), data structures and algorithms, object-oriented design, Docker, GitHub Actions (CI/CD), Linux, unit testing (Pytest, Jest), Git, uv

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)